



GASP-NE interim report on UO PM2.5 data in Newcastle City Centre

Axa-Maria Laaperi, Dr Aleksandra Svalova, Dr Graeme Sarson, Prof
Anvar Shukurov

School of Mathematics, Statistics and Physics
Newcastle University

July 2026

1 Executive Summary

The Urban Observatory (UO) has a collection of precision MONITOR sensors. For use in a multi-sensor spatial model as planned for our proposed predictive online tool, we require as many sensors as possible and therefore verification that the UO MONITOR sensor data agree with that from DEFRA and AURN sensors is needed.

For the purposes of this report we will use the following acronyms:

- 2 DEFRA-owned and approved AURN sensors (DEFRA)
- 5 Locally managed automatic monitoring sensors (Local)
- 5 UO precision MONITOR sensors (UO-Mon)
- UO Mesh sensors (UO-Mesh)

We have highlighted the following as areas for priority action/attention:

- To validate the UO-Mon sensors we compared them to their closest DEFRA/Local sensors; all UO-Mon sensors compared had very good agreement with their closest DEFRA/Local sensor.
- Some UO-Mon sensors were compared with further (in distance) DEFRA/Local sensors in order to compare all DEFRA/Local sensors. We found an increase in distance to change the relationship between sensors to resemble an exponential, as expected, as distance between sensors increased their relationship also changed.
- We will proceed with using the UO-Mon sensors only from the Urban Observatory.
- Some UO-Mon sensors have faulty wind direction and speed data, shown as constant over the course of a day, week and month. We have taken some decommissioned sensor data from 2025 to produce analysis.
- DEFRA Newcastle Centre sensor is systematically rounding PM2.5 values to nearest integer.

The report is structured as follows:

Section 2 shows the locations of all DEFRA, Local and UO-Mon sensors around the Newcastle upon Tyne area that are used within this report on air quality variables (PM2.5, NOx and NO2 concentrations).

Section 3 shows comparisons of PM2.5 concentrations from closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors and closest DEFRA/Local sensors to UO-Mon sensors. March 2026 reasonably demonstrates a longer time period where these sensors could differ in 15min - 1hr intervalled collection. We chose a month that was relatively recent with minimal missing sensors and values. Caveat: there is a short time period between 19th-21st where data is missing from a *DEFRA sensors*.

Section 4 shows comparisons of PM2.5 concentrations to NOx and NO2 concentrations from same sensors for DEFRA, Local and UO-Mon sensors. We chose a shorter time scale for this

analysis, 30th March - 5th April 2026, as NO_x and NO₂ concentrations degrade very fast, so only a short time scale is appropriate for their analysis.

Note:

- We have further analysis on wind speed/direction, temperature and altitude relationships with PM_{2.5} that is not outlined in this report. We also have comparisons with UO-Mesh sensors.
- When examining wind effects, comparing speed and direction to concentration in 2026 was not possible as stated in the above reasoning. With 2025 data, we take old, decommissioned sensors and perform similar analysis with closest UO-Mon sensors and Local sensors. DEFRA sensors were not considered as they are situated further away.

Table of contents

1	Executive Summary	2
2	Sensor locations	5
3	Nearby sensor comparisons	6
4	NOx and NO2 comparisons	10
4.1	DEFRA sensors	10
4.2	Local sensors	11
4.3	UO-Mon sensors	11
5	Further Work	12

2 Sensor locations

The three maps in Figure 1 show the closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors, closest DEFRA/Local sensors to UO-Mon sensors. These pairs will be used to analyse correlations between closest sensors and validate UO-Mon sensors. The distances (km) between these pairs are stated in Section 3 correlational analysis.



Figure 1: Maps of (left to right) closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors, closest DEFRA/Local sensors to UO-Mon sensors, and closest UO-Mon sensors to UO-Mesh sensors.

3 Nearby sensor comparisons

We used March 2026 to look at correlation comparisons between the closest DEFRA/Local sensors to UO-Mon sensors. March was chosen as the dataset had reliable consistent data, i.e. minimal missing values and missing sensor data however there was still a portion of the data missing around the 19th-21st from the DEFRA sensors.

Figure 2 shows closest UO-Mon sensors to DEFRA/Local sensors, scatter plots and time series, where only the closest UO-Mon sensor is included for each DEFRA/Local sensor. The aim of this exercise is to validate the UO-Mon sensor data and from both sets of plots we see very good agreement between the sensors. Figure 3 shows a similar comparison, but the UO-Mon sensors under consideration (UO-Mon Gateshead Tyne Bridge and UO-Mon RVI) have already been compared to other DEFRA/Local sensors in Figure 2 that are much closer, we see worse agreement for two of the sensor comparisons here which is expected due to the distance. A summary table of correlational information and distances between sensors is given in Table 1 from which we see high correlation between all closest sensor pairs. We use Pearson's correlation coefficients. Pearson's evaluates linear relationships and proportional change between variables. Figure 4 also demonstrates the relationship between distance and the correlation coefficients, we see that in general lower correlations do correspond to longer distances.

Other relationship (non-linear) might also exist; we have examined other external factors that are not included in this report but are available if they are of interest.

In Figure 3, we also examined the agreement between Local-Gateshead A1 Dunston and Local-Gateshead Tyne Bridge since we have a comparison with UO-Mon Gateshead Tyne Bridge (in the same location). This way we can deduce if the disagreement is indeed related to distance and excluding sensor related issues. We see a similar nonlinear relationship occurring between these two sensors as with the comparison between Local-Gateshead A1 Dunston and UO-Mon Gateshead Tyne Bridge which indicates this non linear relationship is mainly due to distance between sensors.

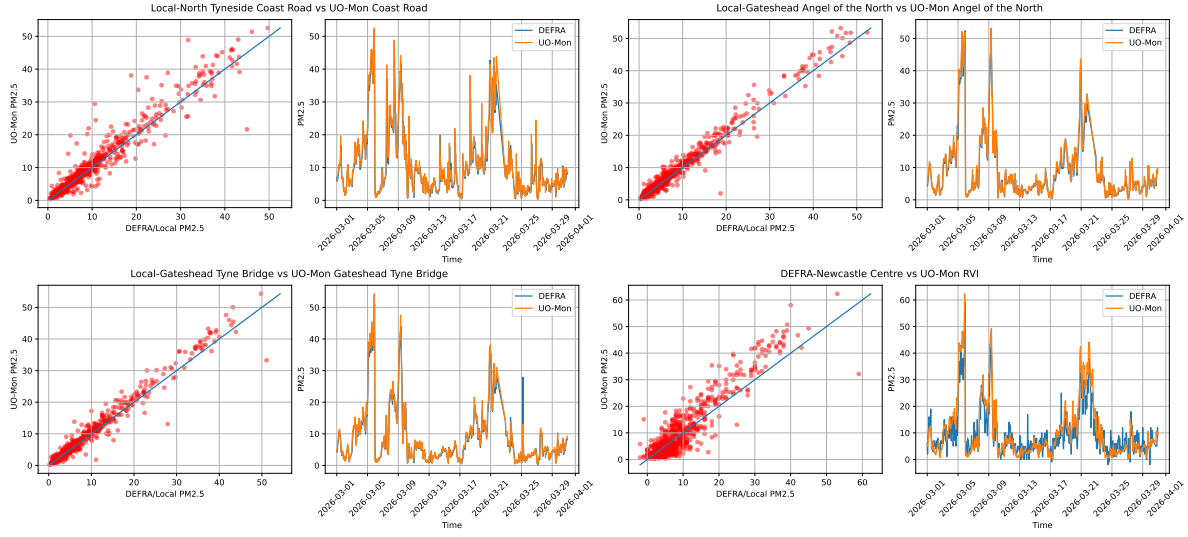


Figure 2: Comparison of PM_{2.5} concentrations measured by each DEFRA/Local sensor and its closest UO-Mon sensor over the study period. Only the closest UO-Mon sensor is included for each DEFRA/Local sensor. We see very good agreement between all sensor comparisons.

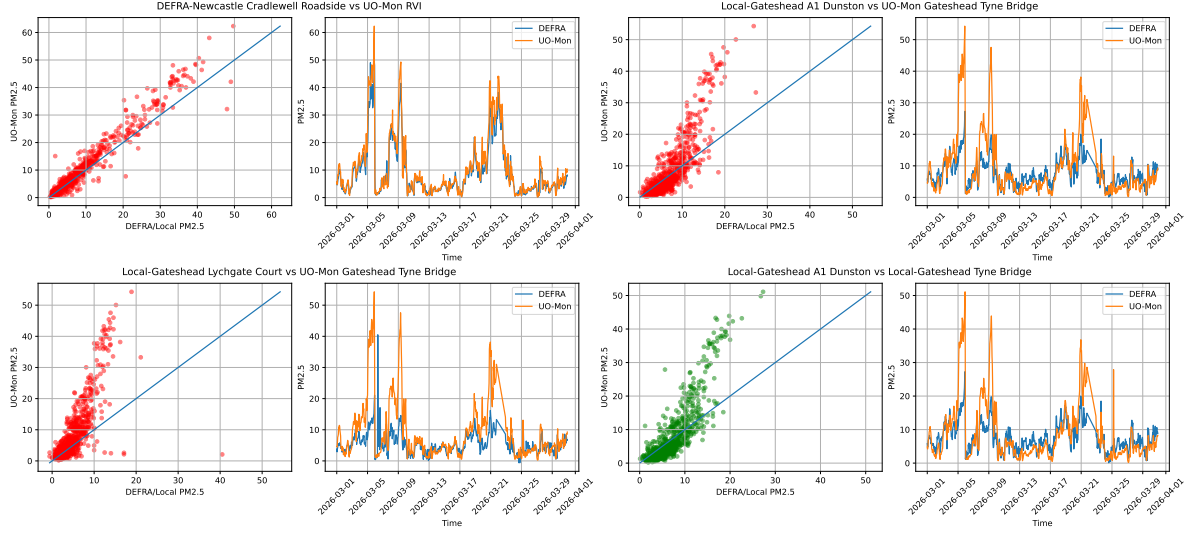


Figure 3: Comparison of PM2.5 concentrations measured by DEFRA/Local sensors and additional nearby UO-Mon sensors over the study period. These comparisons exclude the nearest UO-Mon sensor shown in Figure 2 and illustrate relationships with other matched UO-Mon sensors. We see worse agreement with some non linear relationships which is expected due to the distance. The final plot (green scatter) shows PM2.5 concentrations measured by Local-Gateshead A1 Dunston and Local-Gateshead Tyne Bridge. We expect to see a similar non linear relationship to that comparing Local-Gateshead A1 Dunston and UO-Mon Gateshead Tyne Bridge.

DEFRA/Local	UO-Mon	Distance km	Pearsons coeff
Local-Gateshead Angel of the North	UO-Mon Angel of the North	0.003	0.986
Local-Gateshead Tyne Bridge	UO-Mon Gateshead Tyne Bridge	0.006	0.983
DEFRA-Newcastle Cradlewell Roadside	UO-Mon RVI	1.643	0.974
Local-North Tyneside Coast Road	UO-Mon Coast Road	0.002	0.961
DEFRA-Newcastle Centre	UO-Mon RVI	0.468	0.926
Local-Gateshead A1 Dunston	UO-Mon Gateshead Tyne Bridge	3.37	0.846
Local-Gateshead Lychgate Court	UO-Mon Gateshead Tyne Bridge	0.572	0.741

DEFRA/Local	UO-Mon	Distance km	Pearsons coeff
-------------	--------	-------------	----------------

Table 1: Distances between sensors in km, Pearson’s for UO-Mon and closest DEFRA/Local sensors PM2.5 concentrations over long time interval. High correlational coefficients for all close sensors.

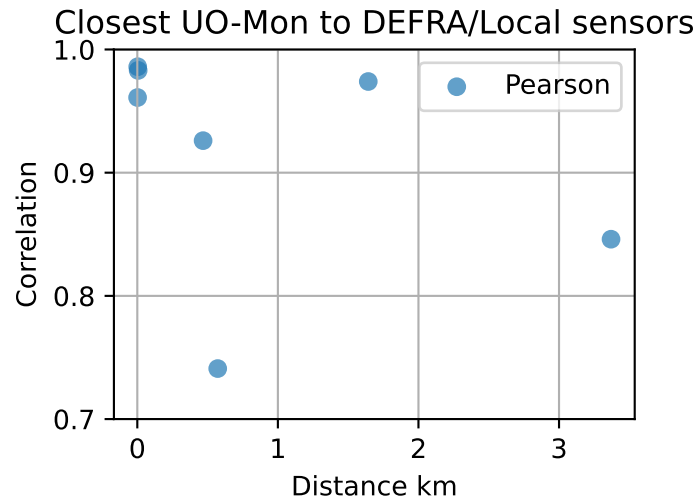


Figure 4: Correlation coefficients of each sensor pair from Figure 2 and Table 1 with their distance (km).

4 NOx and NO2 comparisons

We looked at the last week (30th March-5th April) of March 2026 to examine correlational comparisons of PM2.5 and NOx/NO2 concentrations in the same sensors for all DEFRA, Local and UO-Mon sensors. We chose nitric emissions as a proxy of traffic influence to check whether this has an effect on PM2.5. A shorter time frame was chosen due to nitric particle decay occurring much faster than that of PM2.5.

4.1 DEFRA sensors

First NOx and NO2 concentration comparison in the DEFRA sensors to discern if PM2.5 has any traffic dependency in the same way that NO2 and NOx have. In Figure 5 scatter plots, we see very clearly here the systematic categorisation of the DEFRA Newcastle Centre sensor. Low Pearson's correlation coefficient can be seen for all cases in Figure 5.

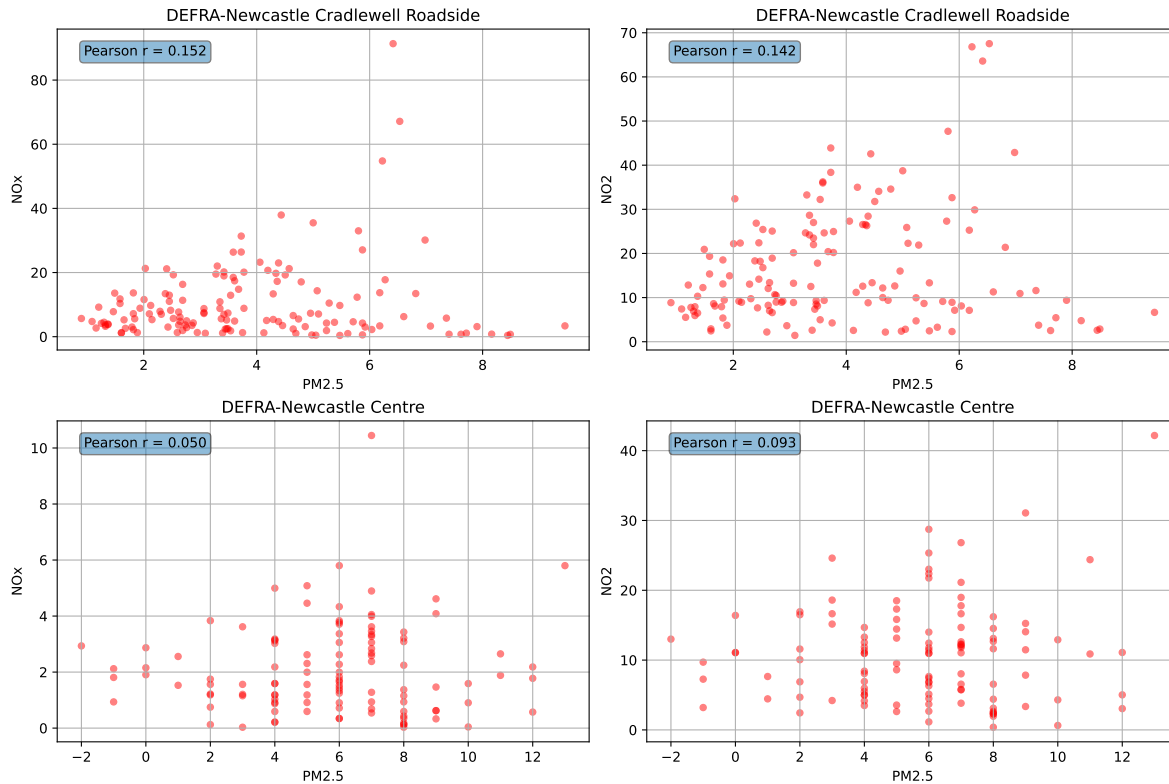


Figure 5: Comparison of PM2.5 and NOx and NO2 concentrations for DEFRA sensors over short time interval.

4.2 Local sensors

In Figure 6, we can see scatter plots for NO₂ comparison with PM_{2.5} concentration for the Local sensors, which do not collect NO_x data. Moderate correlation coefficients can be seen for each case which suggest there may be some traffic dependence or possibly other factors like temperature affecting PM_{2.5} concentrations.

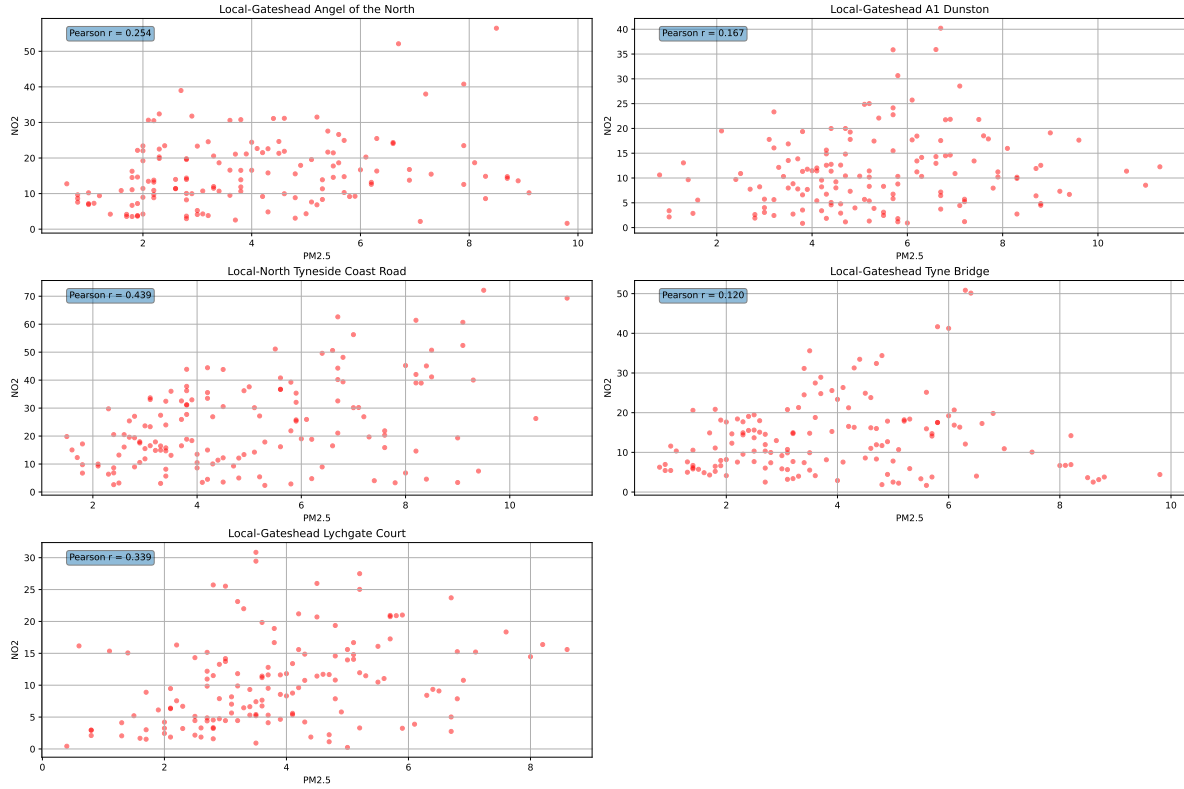


Figure 6: Comparisons of Local sensor PM_{2.5} and NO₂ concentrations over short time interval.

4.3 UO-Mon sensors

In Figure 7, scatter plots for NO_x comparison with PM_{2.5} concentration for the UO-Mon sensors. We see that some of the UO-Mon sensors show moderate positive correlation between PM_{2.5} and NO_x, but the other half show little to no correlation.

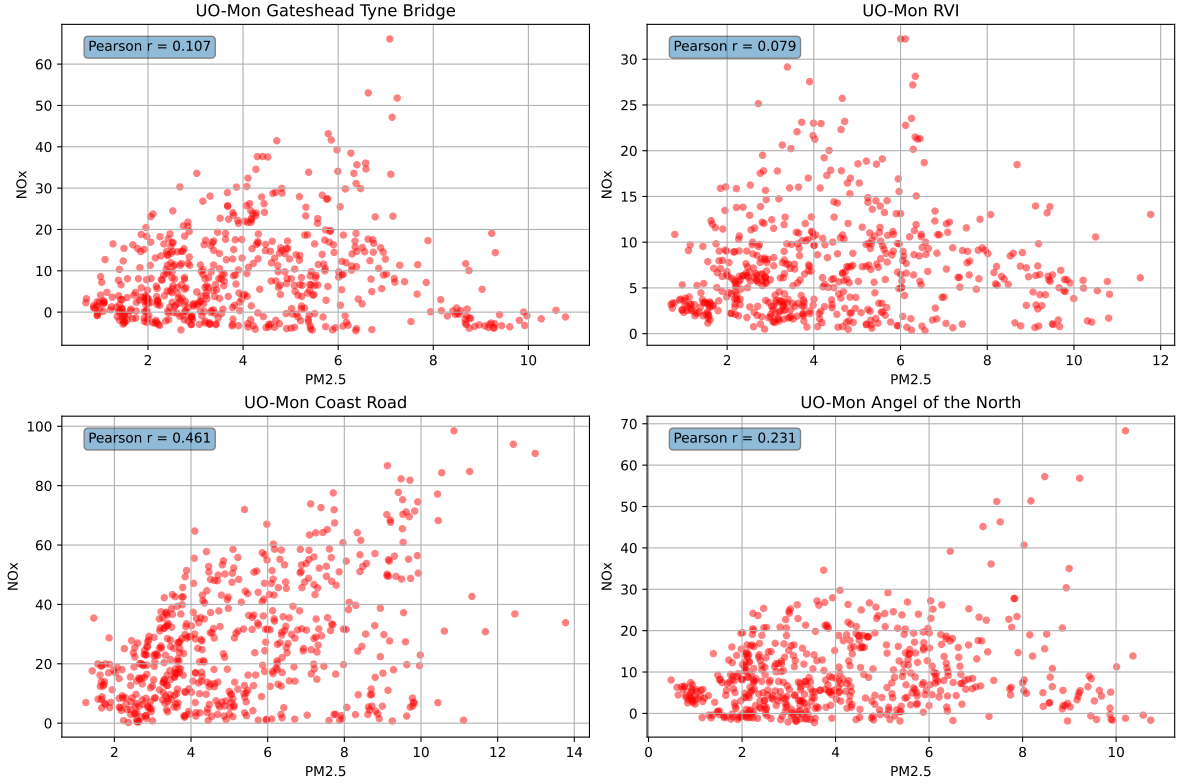


Figure 7: Comparisons of UO-Mon sensor PM2.5 and NOx concentrations over short time interval.

5 Further Work

The analyses presented in this report demonstrates that the DEFRA, Local and UO-Mon sensors exhibit sufficient agreement to support their combined use in further modelling. We investigated the influence of external factors, including wind conditions, temperature and sensor altitude, to provide further context for the observed PM2.5 measurements. Based on this, we will proceed using the DEFRA, Local and UO-Mon sensors to develop a multi-sensor spatial Gaussian Process (GP) model, which will form the basis of the proposed predictive online tool.